

"Kinetics of plasticity regulate burst frequency in a *Dendronotus iris* Swim CPG Model"

James Scully,^{1, a)} Akira Sakurai,^{2, b)} Paul Katz,^{1, c)} and Andrey L. Shilnikov^{3, d)}

¹⁾Neuroscience Institute, Georgia State University, 100 Piedmont Ave., Atlanta, GA 30303, USA.

²⁾Department of Mathematics & Statistics, Georgia State University, 25 Park Pl., Atlanta, GA 30303, USA.

³⁾Neuroscience Institute and Department of Mathematics & Statistics, Georgia State University, 100 Piedmont Ave., Atlanta, GA 30303, USA.

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The swim central pattern generator (CPG) of *Dendronotus iris* produces robust rhythmic activity despite variability in its components and external perturbations. This study examines the contributions of slow synaptic dynamics and the coexistence of multiple oscillatory pathways to the resilience of this network. We constructed a four-cell model based on experimental data, incorporating slow synaptic processes and fast-slow decomposition. Preliminary findings indicate that robustness emerges from the combined influence of multiple oscillatory mechanisms rather than reliance on a single dominant pathway. This model provides a framework for understanding adaptive motor rhythms and their application to biological and synthetic systems.

Keywords: locomotion, modeling, half-center oscillator, hco, excitatory, inhibitory, rhythmic, oscillations, bursting, neurons, subnetworks, synaptic, metabotropic, network, hysteresis, neuroscience, fast-slow, adaptability, robustness, resilience

I. INTRODUCTION

Animal locomotion emerges from neural circuits that must remain effective despite intrinsic variability, changing body size, and fluctuating environmental conditions. Early work on locusts demonstrated “fictive locomotion,” revealing that rhythmic motor output can persist without sensory feedback¹. This led to the identification of autonomous, rhythmogenic circuits known as central pattern generators (CPGs) that drive locomotion in both invertebrates and vertebrates^{2,3}.

Because locomotion underpins critical behaviors like foraging and predator avoidance, CPGs evolve under selective pressure to adapt to environmental variables like temperature and maintain appropriate scaling among body size, locomotion speed, and rhythmic frequency⁴. These circuits integrate sensory feedback and descending control signals, relying on dynamical mechanisms across multiple timescales: spikes at milliseconds, bursts shaped over seconds, and homeostatic feedback stabilizing circuits over minutes to days².

Prior research has shown that ion channel degeneracy allows broad variability in conductances among genetically identical individuals to yield similar functional states^{5,6}, with homeostatic feedback linked to intracellular calcium stabilizing neuronal activity⁷. However, this does not explain how synapses and cells coordinate to produce reliable bursting across a range of frequencies. We propose that controllability, the ability to produce diverse behaviors by modulating

a control variable, is another key factor. In this study, we demonstrate how the kinetic properties of synaptic enhancement allow burst frequency control in the *Dendronotus iris* swim CPG¹⁵.

Initially driven by evolutionary questions about locomotor diversity within its clade—including dorsal-ventral swimmers, left-right swimmers, and non-swimmers¹⁶—our research on the *Dendronotus iris* swim CPG uncovered a novel role for neuroplasticity. Our modeling indicates that the temporal properties of synaptic enhancement, rather than its magnitude or static strength, confer precise control over the network’s bursting frequency and pattern [neuroplasticity citation]. Specifically, the kinetics of synaptic enhancement provide network-level adaptability in this system¹⁵.

Looking ahead, this work highlights the critical role of synaptic plasticity kinetics in governing neural circuit dynamics [neuroplasticity citation]. These properties may serve as a general mechanism for controllability across diverse systems, though the biological processes regulating them remain underexplored. Synaptic enhancement spans facilitation, augmentation, and post-tetanic potentiation, with augmentation matching the timescale of bursting in the *Dendronotus iris* CPG [neuroplasticity citation]. This could involve co-regulation of enhancement forms or selective modulation of augmentation, possibly via synaptotagmin isoforms [neuroplasticity citation].

The paper is organized as follows: First, we describe the swim CPG network assembly, detailing neuron and synaptic interaction modeling. Second, we validate the network by comparing simulated voltage traces with empirical data from laboratory experiments involving current injections and perturbations. Finally, we explore how burst frequency is precisely controlled through synaptic enhancement kinetics, demonstrating the pivotal role of intermediate timescale dy-

^{a)}Electronic mail: jamesjscully@gmail.com

^{b)}Electronic mail: akira.gsu@gmail.com

^{c)}Electronic mail: pkatz@umass.edu

^{d)}Electronic mail: ashilnikov@gsu.edu

namics in adaptive locomotion.

II. RESULTS: BODY SIZE SWIM PATTERNS, FACT CHECKS OR VERIFICATION OF THE CPG MODELS

- A. Resilience to pulse perturbations. Fact check list I
- B. Resilience to pulse perturbations. Fact check II
- C. Si1 controls swim initiation and termination. Fact check III
- D. Role of neuromodulation and its modeling. Burst frequency control

III. METHODS: CELLULAR AND SYNAPTIC MODELS

IV. CPG ASSEMBLY LINE

The central objective of this study is to develop and validate a computational model of the *Dendronotus iris* swim central pattern generator (CPG) that enhances network robustness through a four-cell circuit. This was achieved by constructing the model stepwise, beginning with well-characterized two-cell subnetworks and integrating them into a larger network. A systematic approach ensured biological plausibility, dynamic stability, and alignment with empirical data. Detailed analysis of the neuronal, synaptic, and subnetwork models, including fast-slow decomposition and two-cell oscillatory subnetwork construction, is provided in¹⁸. We summarize the essential details here for convenience.

Neuronal Model

We adapted a conductance-based swim interneuron (SiN) model from Plant's 1981 framework^{18,22}, modifying it with an additional fast I_h current to represent the cellular components of the *Dendronotus iris* CPG. The governing equation is:

$$C_m \frac{dV}{dt} = -I_{Na} - I_{Ca} - I_K - I_{leak} - I_h - I_{KCa}, \quad (1)$$

where C_m ($1 \mu\text{F}/\text{cm}^2$) is the membrane capacitance, V is the membrane voltage, and ionic currents include:

- **Fast inward current:** A combined sodium-calcium current with instantaneous activation and slow inactivation, akin to Hodgkin-Huxley, driving action potentials.
- **Slow inward current:** A TTX-resistant current with slow activation, contributing to bursting periodicity.
- **Fast outward current:** A potassium current, Hodgkin-Huxley-like, for repolarization.
- **Slow outward current:** A calcium-sensitive potassium current, tied to slow intracellular calcium buildup, shaping burst duration.

- **Leak current:** A passive inward current for baseline dynamics.
- **Fast hyperpolarization-activated current:** I_h , added to enhance dynamics at hyperpolarized voltages.

The full description of the model equations are given in Ref.¹⁸, see also the Appendix I below.

Slow variables, intracellular calcium concentration $[\text{Ca}]$ and gating variable x , evolve as:

$$\frac{dx}{dt} = \frac{x_\infty(V) - x}{\tau_x}, \quad \frac{d[\text{Ca}]}{dt} = \rho (K_c x (E_{Ca} - V + \Delta[\text{Ca}]) - [\text{Ca}]), \quad (2)$$

where $x_\infty(V)$ and τ_x (ms) are voltage-dependent activation and time constants, and ρ (ms^{-1}) and K_c (mM^{-1}) control calcium influx and decay.

The parameter $\Delta[\text{Ca}]$ controls the intrinsic excitability of the neuron. Decreasing it is equivalent to increasing the cell's excitability. The parameter Δ_x changes how the cell responds qualitatively to synaptic drive. When Δ_x is 0, the model has an endogenous bursting regime. Decreasing Δ_x to -4 mV eliminates this endogenous bursting.

Synaptic Models

Synaptic dynamics are modelled using a phenomenological approach detailed in¹⁸, which balances realism and tractability. Two synapse types were implemented:

1. **Fast synapses:** Modeled with a gating variable S that rises quickly with presynaptic voltage and decays at a fixed rate ($\alpha = 10 \text{ ms}^{-1}$, $\beta = 5 \text{ ms}^{-1}$), following the α -synapse formulation²³.
2. **Slow synapses:** Include a nonlinear term for synaptic accumulation (minima above zero) and high-pass filtering of low-frequency inputs.

Full equations are in¹⁸ or the appendix.

Four-Cell Network Assembly

We constructed the four-cell network using validated two-cell or pairwise subnetworks such as half-center oscillators (HCOs) and excitatory-inhibitory (EI) modules introduced in Ref.¹⁸.

Synaptic kinetics for mutual inhibition matched HCO results, and E/I interactions were similarly aligned.

We preserved cellular parameters and synaptic kinetics, adjusting only the endogenous excitability of Si3 (swim interneuron 3) by shifting its voltage threshold $\Delta[\text{Ca}]$ by 5mV to enhance spiking. Calibration relied on symmetrically tuning synaptic conductances, following these steps:

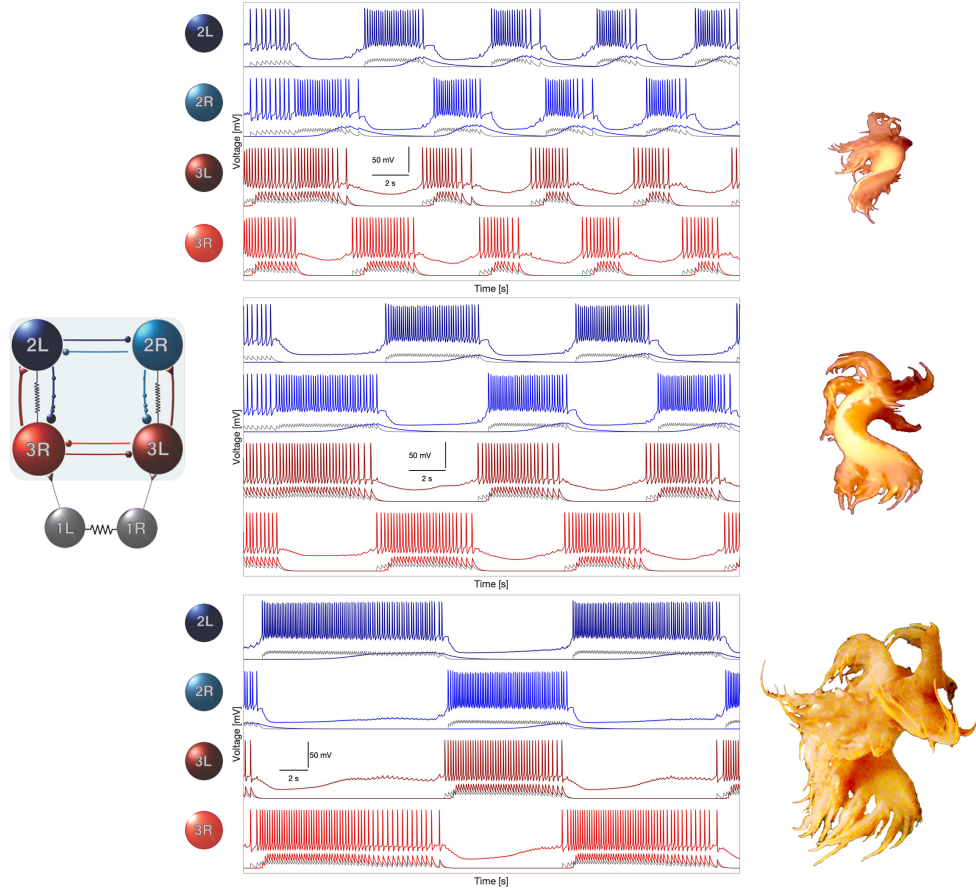


FIG. 1. A three-panel collage illustrating diverse swim patterns generated by the swim CPG circuit, adapting to the maturing size of *Dendronotus iris*. (A) Swim CPG circuitry featuring core interneurons Si2L/R and Si3L/R, driven by the sensory interneurons Si1s. (B) Voltage traces (top) and neurotransmitter release probabilities for the four core swim CPG interneurons (SiN). (C) As *Dendronotus Iris* grows from 30g to 300g over its lifespan, as shown in the voltage traces in panel B, its swim period slows down from 2–3s in juveniles to up to 12–14 seconds in older adults, see the voltage traces in panel B.

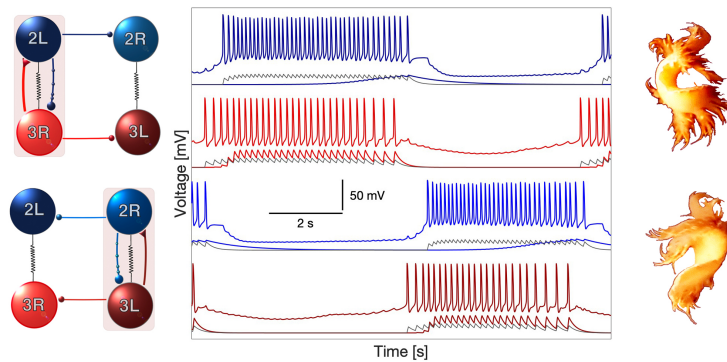


FIG. 2. Anti-phase left-right episodes within a swim pattern, produced by the *Dendronotus iris* CPG under control conditions, arise from twisted pairwise modules ($Si3R \leftrightarrow Si2L$ and $Si3L \leftrightarrow Si2R$) that are unidirectionally coupled via excitatory (▼) and inhibitory (●) synapses. These modules alternate in initiating the onset of network bursting, an effect further reinforced by two reciprocally inhibitory HSOs, consisting respectively of Si3R–Si3L and Si2R–Si2L pairs. Compare with the original Fig.1 from the experimental study¹³.

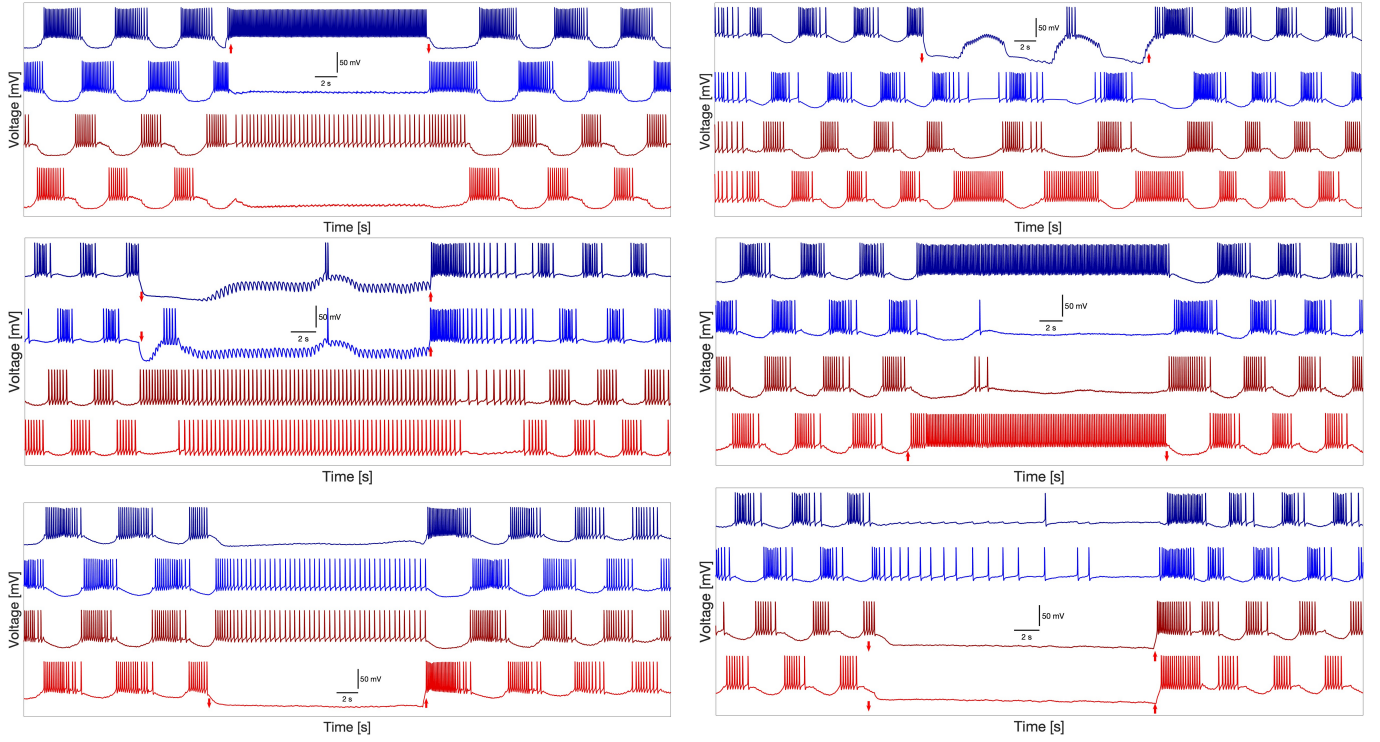


FIG. 3. Comparison of specific, distinct responses of the swim CPG to specific perturbations using hyper- and depolarized current pulses (indicated by red vertical arrows) injected into the targeted interneurons let us validate the CPG model against its biological original and demonstrate the resilience of both to restore and proceed with the normal swim patterns. Fact check list I: to compare these six modeling panels with the corresponding experimental originals: Figs. 5A–C and E–G, resp. from Ref.¹⁵. For the empirical resilience evidence see also Fig. 5 in Ref.¹⁴; for the PIR de-facto proof see Fig. 5 in the experimental Ref.¹³.

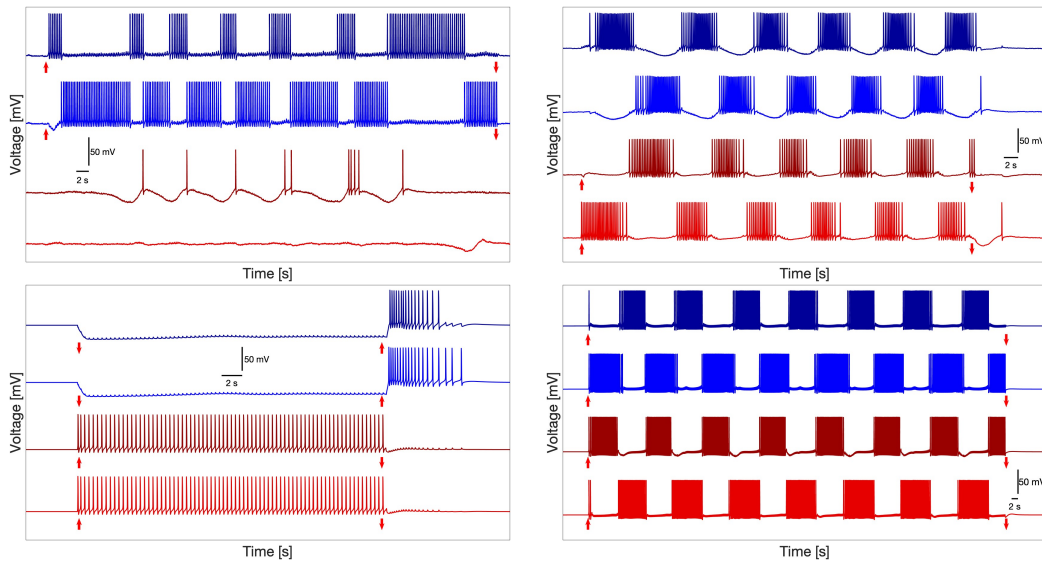


FIG. 4. Similar responses of the swim CPG in control swim to various temporal perturbations using hyper- and depolarized current pulses (indicated by red vertical arrows) applied to the targeted interneurons let us validate the CPG model against its biological originals Fact check list II: to compare these four modeling panels with the corresponding experimental originals: Figs. 7Ai–Di from Ref.¹⁵

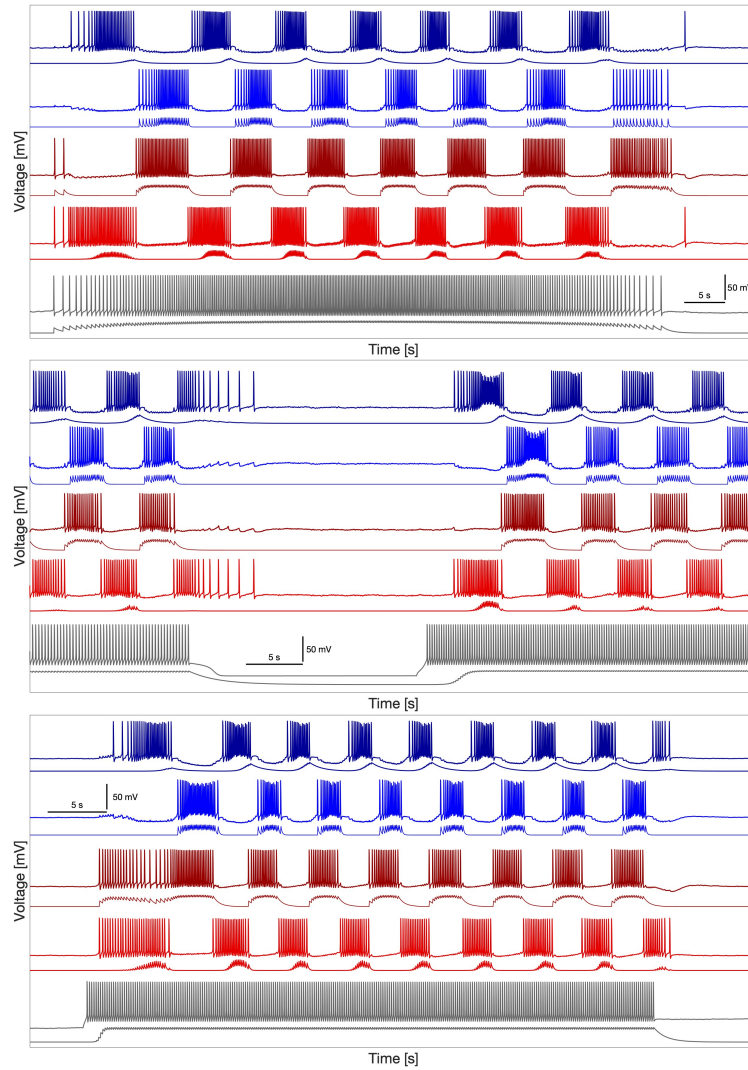


FIG. 5. Interneuron Si1 drives Si3s with excitation, and hence initiate and terminate the whole rhythmically bursting CPG [without neuromodulation]. Fact check list III: to compare i) panel A with its experimental original – Figs. 2A from Ref. ¹²; as well as ii) panels B and C with the corresponding originals – Figs. 3A and 3B ¹², resp.

1. **Configuration.** Substituted pre- and postsynaptic voltage variables to reflect the known *Dendronotus iris* circuit ¹⁵.
2. **Initialization.** Set all cells (Si1–Si4) to a quiescent state and synaptic conductances to zero.
3. **Drive.** Applied synaptic drive to Si3 until the excitatory postsynaptic S accumulated with minima above 0.1, saturating over 0.5–3 s (1/4 to 1/2 the burst period of 2–12s, varying with animal size).
4. **Excitation.** Increased excitatory conductance from Si3 to contralateral Si2 until inhibitory S (Si2-to-Si2) accumulated with minima above 0.1 over 0.5–3 s.
5. **Slow Inhibition.** Raised E/I pair inhibitory conductance until Si3 frequency dropped by 50% over 1 s.
6. **Fast Inhibition.** Elevated inhibitory conductances for both inhibitory HCOs (Si1–Si2 and Si3–Si5) until a stable burst rhythm emerged, visually resembling empirical voltage traces ¹⁵.
7. **Fine-Tuning.** Adjusted excitability slightly (e.g., $\Delta[Ca] < 2$ mV) to set a 30% duty cycle, verified as extrinsic by resetting conductances to zero.

TODO: Are we using Si1 or Iapp for Si3 drive?

A. Validation from voltage perturbation

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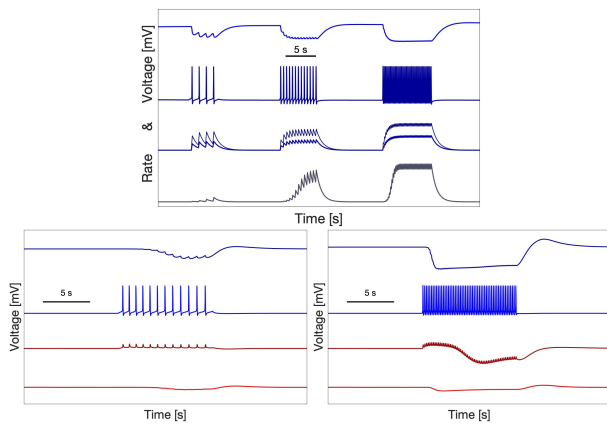


FIG. 6. Fact check IV to compare these panels B and C with the original neurophysiological data – Fig. 7 in Ref.¹³. Three positive current pulses of increasing amplitude are applied to Si2R to calibrate α -synapse and logistic synapse (bottom trace) models and IPSP responses in the counterlateral Si2L (top trace). Two positive current pulses injected into Si2R trigger accumulating IPSP responses in Si2L (top traces) and fast and slow echoes in the Si3L due to fast gap junction and delayed slow inhibition due to the logistic synapse between Si2R and Si3L, see the swim CPG circuitry.

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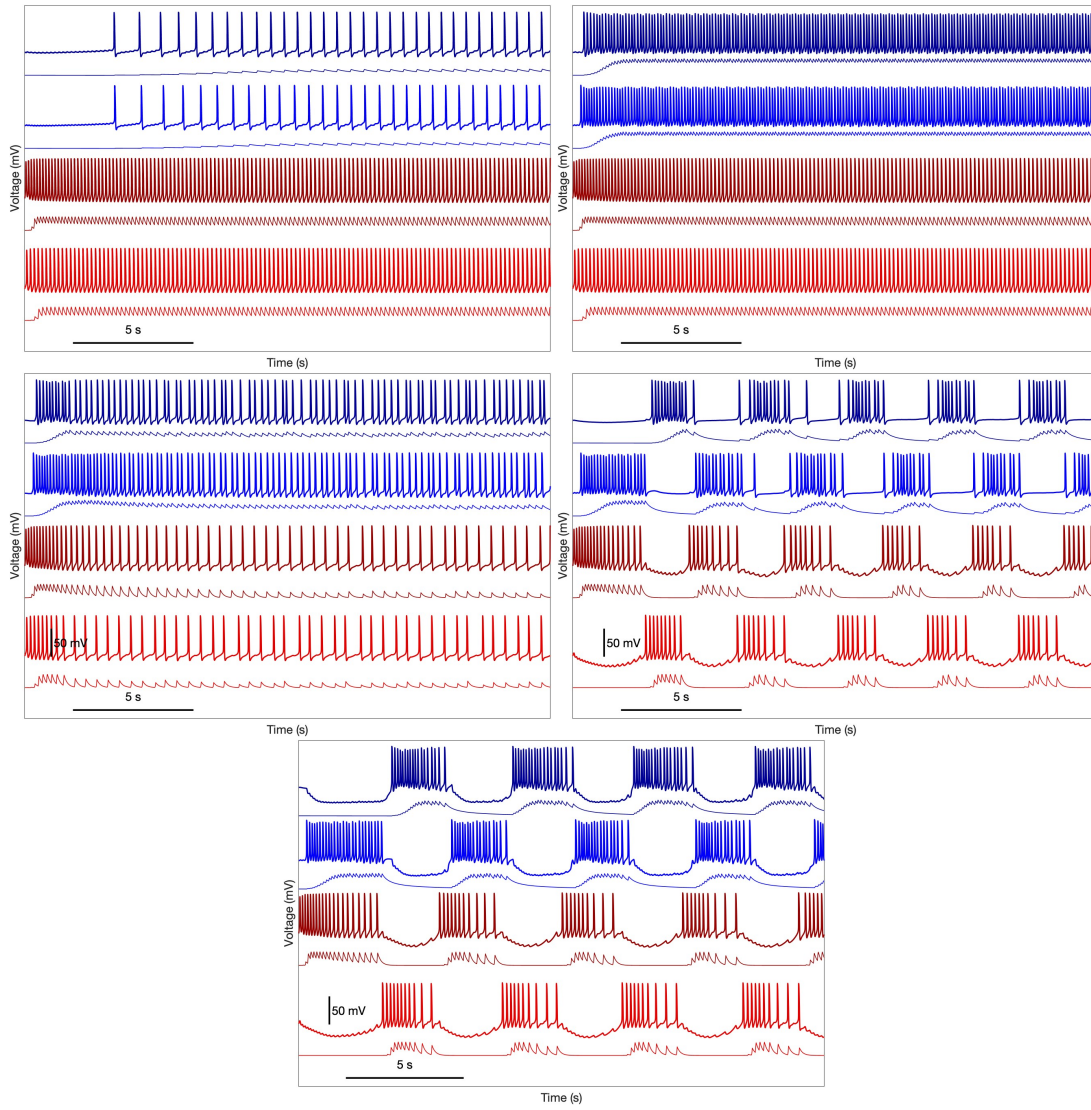


FIG. 7. CPG assembly line in five sequential steps [in arbitrary order]. (A) Uncoupled four interneurons, with Si3's being at higher spike rate due to "excitatory drive" from Si1's. (B) Si3's drive ipsilaterally Si2's with fast excitatory drive. (C) Si2's return inhibitory feedback onto Si3's. (D) Reciprocally inhibitory synapses are turned on between contralateral Si2's, along with gap junctions between Si3's and Si3's, to generate stably the fast network-level bursting in the swim CPG model of *Dendronotus Iris*.